

Fast Archipelago v1.3

One-shot fast agreement with certified slow fallback

Protocol revision and implementation specification

4 October 2026

Abstract

This revision replaces the repeated per-rank fast/recovery construction in v1.2. Each checkpoint has one immutable fast-vote attempt. Recovery certificates supply externally valid initial inputs to a fresh ordinary Archipelago instance. The slow instance runs its own R, A and B phases; recovery never overrides its R result. Fast evidence remains actionable after slow execution starts. The fast/recovery composition argument is given below. End-to-end termination remains conditional on the stated ordinary slow-engine and dissemination obligations; this document does not claim those implementation obligations are already discharged.

1 Model and domains

Fix a checkpoint instance I binding protocol version, session, epoch, predecessor checkpoint and the exact equal-weight committee and fault budgets. There are $n = 3f + 2p + 1$ identities, at most f Byzantine identities, and unforgeable signatures. Correct identities never change their first vote in I . Let

$$F = n - p, \quad Q = n - f - p = 2f + p + 1, \quad P = n - f = 2f + 2p + 1.$$

These are fast/recovery thresholds, not the slow engine's quorum thresholds. Values have a deterministic total order and must satisfy the application validity predicate, including reconstructed payload availability and predecessor binding. A hash alone does not establish application validity.

For the concrete fallback in this revision, choose the lexicographically smallest $3f + 1$ public keys from the fast committee, fixed for I . Define I_s by hashing an independent slow-instance domain, I , f and the ordered selected keys. The slow committee has at most f Byzantine identities, quorum $q_s = 2f + 1$ and witness threshold $w_s = f + 1$. Fast members outside this committee still serve fast votes, recovery proofs and application data. Changing this selection rule requires a protocol version change.

Every signed object binds a message-kind domain, its instance, signer, value and relevant phase/rank/request identifiers. Fast attempt number is always zero; slow rank zero is a distinct namespace. v1.2 signatures are not v1.3 signatures. A slow proposal signature binds its value and slow instance; its attached RC is independently verified. Different valid RC witnesses for the same value are interchangeable. Subsequent slow phase signatures must bind their phase-specific proof references under the ordinary engine's rules.

2 Finite application admission and initial R evidence

Before releasing FIRST, the RAI binding authenticates a candidate object containing I , its proposer and the complete epoch value. Validation reconstructs exactly $P = n - f$ distinct committee reports,

checks both roots and their signed committee and predecessor context, and recomputes BuildState. A signature alone is not application validation.

A candidate becomes an admissible initial value only with Q distinct signed endorsements of that exact candidate object and proposer. Each correct endorser locks one candidate per $(I, \text{proposer})$ for the process lifetime. Two admission quorums for one proposer intersect in at least $2Q - n = f + 1$ identities; at least one is correct and cannot endorse both. Hence at most one admitted candidate exists per proposer, and there are at most n initial values. Correct proposers must release one validated candidate and dissemination must make its payload available. Admission is an additional exchange before the optimistic FIRST attempt; latency measurements must include it when measuring the complete checkpoint operation. Disk/restart safety for endorsement locks is deferred.

For empty-candidate recovery the implementation adds an independent initial max-register exchange, in a domain separate from FIRST and the RC-backed slow instance. An initial R request names an admitted candidate. A correct responder incorporates the request into its retained maximum and signs a response binding I , the request digest and its maximal admitted candidate. An initial R certificate contains Q distinct such responses to the same authenticated request. Verification checks signatures, admitted origins, and that each response is at least the triggering input, and recomputes their maximum. The cache stores only verified certificates; the FIRST snapshot's maximum is never substituted. Candidate-admission quorums witness origin availability in this initial exchange; these are not the later slow engine's R-response witnesses. This specifies the initial max-register refinement used by the application adapter. It does not supply the ordinary slow synchronizer or discharge its termination assumptions.

3 One initial fast attempt

The initial fast R exchange retains the eligible-request/response and witness checks of the fast/recovery front end. A correct replica authenticates a valid initial proposal and releases at most one signed $\text{FIRST}(I, 0, v)$. Later information can change an ordinary R response, but cannot change FIRST. The immutable record must be installed atomically before releasing the signature. There are no FIRST votes at slow ranks, including later slow ranks.

A fast certificate $FC_0(v)$ contains exactly F distinct valid FIRST signatures for v . Receipt of such a certificate permits deciding v at any time, including while the slow engine is running. Correct replicas relay certificates and dependencies; starting the fallback does not disable this listener.

The implementation currently retains locks only in memory, at the user's request. Therefore its safety scope excludes restart with lost locks followed by signing again in the same instance. Durable restart support is deferred.

4 Recovery and admission

Let S contain one validated FIRST record per identity, with $Q \leq m = |S| \leq n$. Define

$$\text{Cand}(S) = \{v : \text{count}_S(v) \geq m - f - p\}.$$

An equivocating identity contributes at most one record to a snapshot. A receiver may nevertheless retain alternate signed records for detecting a fast certificate.

A resolved certificate $RC_0(x)$ contains I , attempt zero, S , x , and one of:

1. Singleton evidence: $\text{Cand}(S) = \{x\}$.

2. Empty evidence: $\text{Cand}(S) = \emptyset$ and independently verified initial ordinary R evidence whose recomputed maximum is x .

An ambiguous candidate set does not resolve: neither taking its maximum nor replacing old first votes is allowed. Missing dependencies are fetched and retried; invalid dependencies do not count. Empty-case R evidence must include all required ancestry and eligibility witnesses, not merely a caller-supplied maximum.

Any member of the slow committee may start the fresh slow instance with an initial proposal x accompanied by $RC_0(x)$. All initial slow proposals, including Byzantine proposals, must satisfy this external validity predicate. Validation is independent of the verifier's local recovery snapshot. A valid certificate does not rewrite that verifier's first vote.

5 Timing and ordinary slow ranks

There is a causal boundary: recovery evidence must exist before its value enters slow consensus. There is no wait for a timeout, failed-fast proof, all fast votes, or abandonment of the fast attempt. Slow work and delivery of late initial fast evidence therefore overlap.

The slow instance starts with fresh responder, witness and synchronizer state. Its ordinary rank structure is:

1. R: exchange legal requests, their witnesses and eligible responses; select the ordinary R result r from a valid slow quorum and retained local R state. When the retained maximum exceeds the quorum maximum, its signed record and origin witnesses must accompany the selection proof.
2. A: propose r with its slow R evidence. Verify ordinary A request, response and complementing-witness rules; produce the ordinary A result.
3. B: propose that A result with its proof; verify B evidence and either commit or adopt according to the ordinary B evaluator.
4. On adoption, advance via the ordinary synchronizer and legal preceding-B ancestry. Do not introduce a new fast attempt or replace the next R result by RC.

Request collection barriers, response updates, W admission and Q eligibility are internal rules of the chosen ordinary engine, not interchangeable messages or additional consensus decisions. Their exact source-faithful implementation and recursive proof validation remain required work.

In particular, if the initial external input is v but slow R validly selects w , A must propose w . The initial $RC_0(v)$ is not permission to override R. Every value introduced later must have ordinary legal ancestry back to an externally valid initial input. Neither an old fast certificate nor an arbitrary hash substitutes for preceding slow B evidence.

6 Safety of the composition

Fast uniqueness. Two fast signer sets intersect in at least $2F - n = n - 2p = 3f + 1 > f$ identities. Some intersecting identity is correct and cannot first-vote for two values. Thus conflicting fast certificates cannot exist.

Hidden-fast preservation. Suppose $FC_0(v)$ exists, even if initially hidden. At least $F - f$ correct identities first-voted v . A snapshot of size m includes at least

$$(F - f) + m - n = m - f - p$$

of them. Hence $v \in \text{Cand}(S)$ for every valid recovery snapshot. An empty certificate is impossible, and every resolved singleton certificate selects v . This also applies to a fast certificate assembled later: a correct identity that first-voted another value cannot subsequently join it for v .

Fast/slow agreement. Assume the slow engine enforces external validity on every initial input, preserves legal ancestry, satisfies agreement, and decides v when every externally valid input equals v . If $FC_0(v)$ exists, hidden-fast preservation restricts every admissible slow input to v , so every slow decision is v . Together with fast uniqueness and slow agreement, decisions agree across paths. These are explicit requirements on the ordinary engine, not properties established by the current pure A/B helper functions alone.

7 Recovery progress and conditional termination

At $m \geq P$, two candidates would require at least $2(m - f - p) > m$ votes, since $m \geq 2f + 2p + 1$. Therefore there is at most one candidate. This establishes resolvability of a sufficiently large snapshot; in the empty case it still requires available, independently valid initial R evidence.

Assume reliable eventual dissemination of correct initial FIRST records and required payloads/proofs, eventual production of a valid initial R certificate when needed, and an ordinary slow engine satisfying termination under the execution's scheduling and synchronizer assumptions. Once P correct identities participate, recovery resolves; correct slow members receive admissible proposals and the slow engine terminates, or a fast certificate already decides the instance. Thus the composition terminates under these premises.

For static temporary suspensions, the selected cohort may be unavailable for a finite interval and later resume; this is not permanent absence. After the last return, the fixed slow committee again has the ordinary at-most- f Byzantine fault model. A proof must show that the slow synchronizer recovers from the arbitrary prefix and reaches its progress condition. Static identities alone do not imply such a suffix if suspension intervals can recur forever. This revision does not claim termination during arbitrary recurring or rotating suspensions, nor does it assume all responses become identical merely because values are ordered.

8 Why the repeated construction was withdrawn

In v1.2, recovery could repeatedly replace an ordinary R result w by v before A, while an adversary retained a privately assembled preceding adoption proof for w . Mutable responder evidence permits earlier certificates to remain valid: a later stronger certificate does not revoke signatures already released. Consequently adoption of v and adoption of w can coexist without any commit. The recurring trace in the accompanying investigation reproduces this gap even with a clean suffix. A request barrier or a single update per rank is not, by itself, a proof that a correct first-vote quorum eventually forms.

The repair removes the repeated override: RC is initial external validity for a fresh slow instance, and ordinary R controls ordinary A at every slow rank. This removes the exhibited recovery-induced cycle. It is not a substitute for proving the slow engine's own agreement, validity, synchronizer and

termination. The former unrestricted adoption-persistence and unconditional composed-liveness claims are not retained.

9 Implementation contract and remaining gates

Phase 1 now enforces attempt-zero fast signing and certificate verification, versioned fast domains, deterministic slow membership/instance binding and signed RC-backed initial slow proposals. Tests exercise rejection before signing, recovery verification, replay isolation and late fast completion after fallback admission. Existing A/B and witness helpers remain pure components.

The component implementation now includes signed slow requests, bounded recursive ancestry checks and an in-memory proposer loop, including retained R selection. Request proofs require weak complementing witnesses; quorum eligibility requires strong witnesses. Component tests cover adoption and subsequent commitment. Mutable responder services and digest-based proof fetch/relay now run in a transport-independent replica. Four-replica simulations cover split inputs, missing certificates and service to a returning replica after decision. Typed-message admission and historical-response replay now have configurable byte budgets; replay also has a count limit and rotates fairly without evicting evidence. These are local accounting controls, not wire framing, total-storage limits or round synchronization. Finite initial candidates now have quorum admission and an independent initial R certificate verifier. Transport codecs, a committee-wide driver and local application validation are component implementations. Still required: admission and application-evidence dissemination, retry scheduling and a round synchronizer, live checkpoint service integration, adversarial timing coverage and Gate B end-to-end checks. Aggregate retained-storage limits and disk persistence are deferred; the current nanospam baseline keeps all nodes online. The one-shot interface tests are not an end-to-end termination proof. Disk persistence remains deferred. No production checkpoint service is installed by this revision.

The editable source is `doc/fast-archipelago/fast-archipelago.tex`; `rebuild.py` compiles the complete document. Historical v1.2 analyses and replacement-page source remain separate investigation artifacts, not normative parts of v1.3.